Explaining Voluntary Carbon Market in 10 Minutes

by Sean Huang

View the interactive version (Power BI Desktop/Service needed)
(https://github.com/YuHsiangHuangSean/Voluntary-Carbon-Market-Dashboard/blob/main/dashboard/voluntary_carbon_market_dashboard.pbix)
View the concise version
(Stay tuned!)
View GitHub
(https://github.com/YuHsiangHuangSean/Voluntary-Carbon-Market-Dashboard.git)

What is the voluntary carbon market?

There are many ways to reduce carbon emissions or remove carbon from the atmosphere. All of which, however, require funding. Voluntary carbon market directs funds from companies interested in solving climate change to projects that reduce/remove carbon.

How does the voluntary carbon market work?

Project developers reduce/remove carbon.
Carbon standards issue and track carbon credits after they are verified.
Buyers purchase carbon credits and fund project developers in doing so.

How is this project useful?

The voluntary carbon market has been debated for years, particularly over its role in climate change mitigation. Some see carbon credits as an essential tool for addressing hard-to-abate emissions, while others worry that they may distract from meaningful decarbonization efforts.

Yet during my thesis research, I found a more fundamental challenge: **Understanding the voluntary carbon market itself is surprisingly difficult**. The market is complex and fragmented, with multiple carbon standards, project types, certification schemes, and overlapping terminology.

By analyzing and visualizing publicly available data and combining them with my thesis findings, this project aims to make the voluntary carbon market more accessible and help users understand its current state and possible future development **in 10 minutes**

Data source:

- MSc thesis:

Huang, Y.H. (2026). Rebuilding legitimacy: Interaction dynamics of natural climate solution credits for European markets [Master's thesis, Utrecht University]. Utrecht University Student Theses Repository. https://studenttheses.uu.nl/handle/20.500.12932/52522.

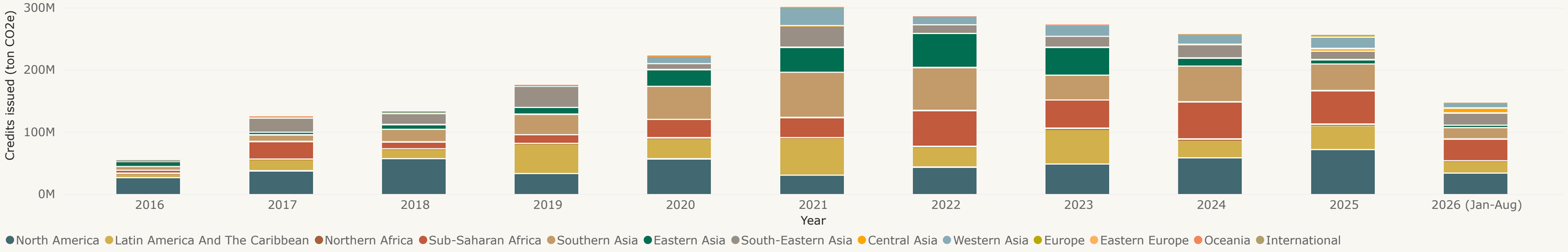
- Carbon project data

Pamela Quartson, Barbara K Haya, Tyler Bernard, Aline Abayo, Xinyun Rong, Ivy S So, Micah Elias. (2026). Voluntary Registry Offsets Database v2026-08, Berkeley Carbon Trading Project, University of California, Berkeley. Retrieved from: https://gspp.berkeley.edu/berkeley-carbon-trading-project/offsets-database.
The current scope of this analysis is projects that issued or retired credits between Jan 2016 and Aug 2026, verified by one of the four major carbon standards (Verra, Gold Standard, ACR, CAR).

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Where are carbon projects taking place?

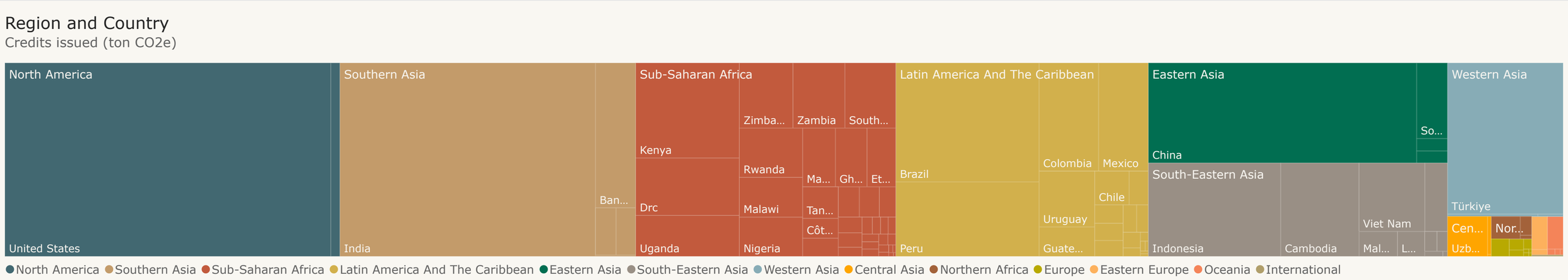


Year

2016 2026

Region

North America | Latin America And The Caribbean | Northern Africa | Sub-Saharan Africa | Southern Asia | Eastern Asia | South-Eastern Asia



Key insights:

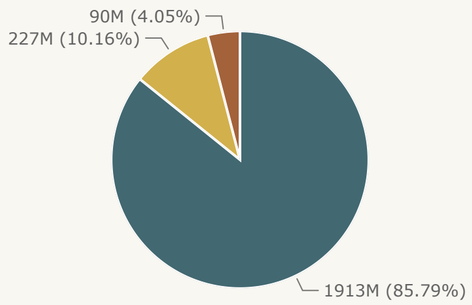
- Voluntary carbon market experienced **a rapid growth** in credit issuance between 2016 and 2021, but has been **slowly shrinking** ever since.
- Before 2021, carbon projects usually take place in **North America** (mostly US), **Southern Asia** (mostly India), and **Latin America**.
- After 2021, **Sub-Saharan Africa becomes an emerging hub** for carbon projects and has surpassed Southern Asia and Latin America.

How are carbon credits generated?

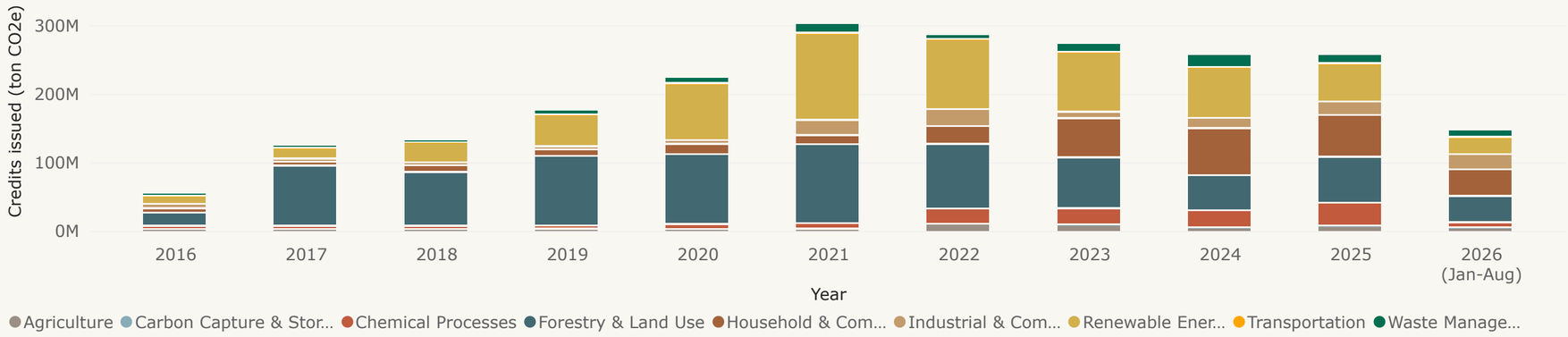
86% are generated by reducing carbon emissions, while 14% involves removing carbon from the atmosphere (carbon removal).

Reduction / Removal

Credit issued (ton CO2e)



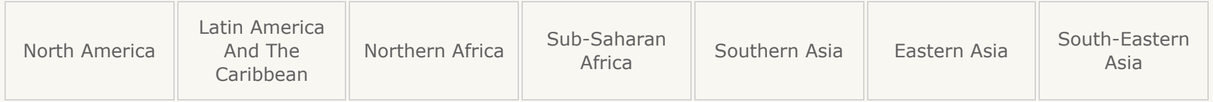
Carbon Project Category



Year

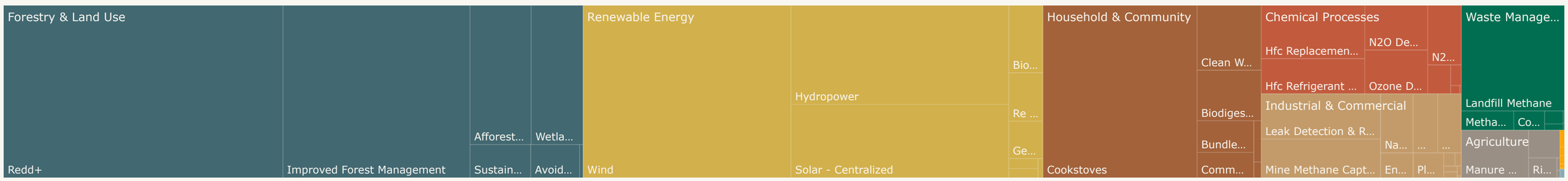


Region



Category Detail

Credits issued (ton CO2e)



Key insights:

- Carbon credits are mostly generated from **forestry** and **renewable energy**. **REDD+** (Reducing Emissions from Deforestation and Forest Degradation) used to be the most popular, particularly in **Latin America**. Since 2023, this is no longer the case.
- **New favorates are wind power and cookstoves**. The former is popular in **Southern Asia**, while the latter is popular in **Sub-Saharan Africa**

Who issues carbon credits?

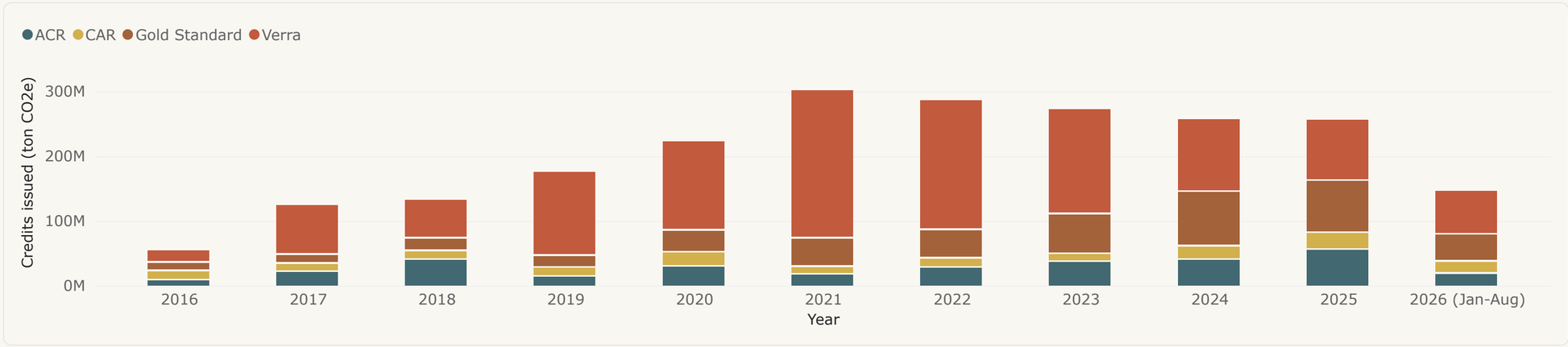
Credits are issued by carbon standards and registered in their registries. This dashboard track four carbon standards, which account for ~89% of the transaction volume[1].

Carbon Standard



Year

2016 2026

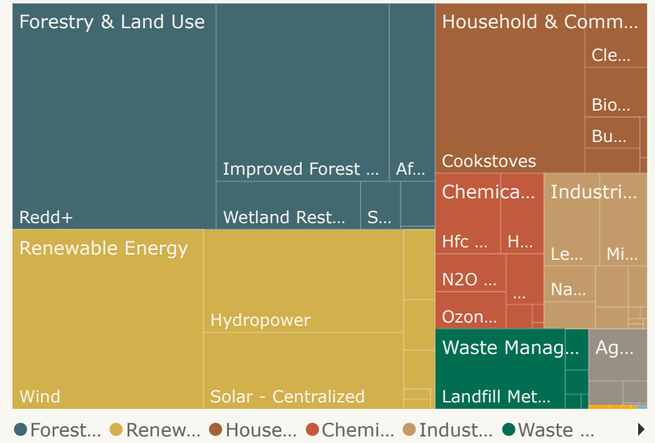


Region	ACR	CAR	Gold Standard	Verra
North America	284804493	162786874	12976	32010321
Latin America And The Caribbean	14420217	19082138	38189174	289696597
Northern Africa			4942864	1800460
Sub-Saharan Africa			147326270	225057630
Southern Asia			116945570	306276780
Eastern Asia			39511125	182239522
South-Eastern Asia	1299990		28304405	176923915
Central Asia			383858	12466979
Western Asia			76174321	55159374
Europe	1216467		202591	3859642
Eastern Europe			1804205	2712625
Oceania			565324	3873889
International			318124	
Total	**301741167**	**181869012**	**454680807**	**1292077734**

Carbon Project Category	ACR	CAR	Gold Standard	Verra
⊞ Agriculture	1857860	9421277	9020066	18545088
⊞ Carbon Capture & Storage	0			303243
⊞ Chemical Processes	96333609	45829404		4627722
⊞ Forestry & Land Use	175509193	90975331	8023734	553674275
⊞ Household & Community			209507143	102440074
⊞ Industrial & Commercial	24723732	8842719	1778830	103843953
⊞ Renewable Energy	1078246		186691022	469238541
⊞ Transportation	302545		415375	746895
⊞ Waste Management	1935982	26800281	39244637	38657943
Total	**301741167**	**181869012**	**454680807**	**1292077734**

Carbon Project Category

Credits issued (ton CO2e)



Key insights:

- **Verra has historically been the dominant carbon standard**. Since 2022, it has faced **increasing competition from Gold Standard,** who stands out for **adopting household projects earlier and maintaining a higher proportion** than other standards.
- Both Verra and Gold Standard have global reach, while **ACR and CAR are primarily active in North and Latin America**.
- Both ACR and CAR initially relied on forestry but have **diversified into chemical processes**.

1.https://www.ecosystemmarketplace.com/publications/2025-state-of-the-voluntary-carbon-market-sovcm

What are the key challenge facing the market?

The accussations from the media and academia about **project quality**:

The Guardian "Revealed: more than 90% of rainforest carbon offsets by biggest certifier are worthless, analysis shows"[2]

What types of projects are accused?

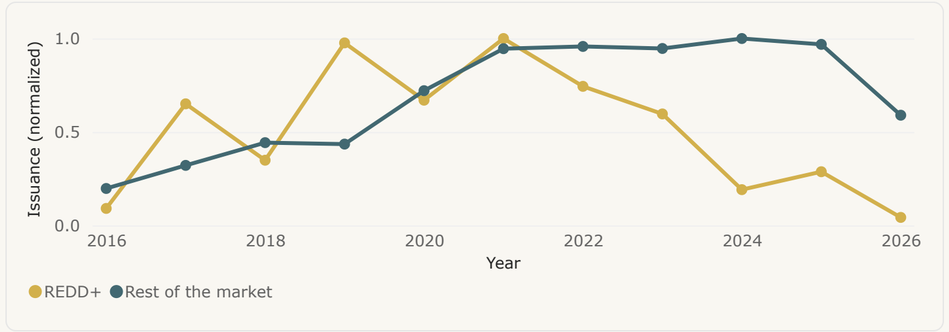
One of the most issued, REDD+ (Reducing Emissions from Deforestation and Forest Degradation).

What are they accused of?

- Over-crediting: Issuing more credits than reduced.
- Providing no co-benefits for local communities as promised.

What are the consequences?

Buyer hesitation towards certain project types, leading to relative declines in issuance.



"All of a sudden everyone just wants reforestation projects, and no one's talking about REDD+ anymore…"
(Business Supplier and Intermediary 1)

The Guardian articles "*brought the market down to earth.*"
(NGO Watchdog 1)

"...after all those controversies with avoidance, it's just something we don't want to touch."
(Business Buyer 1)

Solution: transition from quantity to quality, supported by the usage of additional certification labels

ICVCM CCP

A multi-stakeholder initiative that approves carbon standards and their methodologies

Article 6.2 of the Paris Agreement

UN regulation for nation-to-nation transactions of credits

CORSIA

UN regulation for reducing and offsetting international airlines' emissions

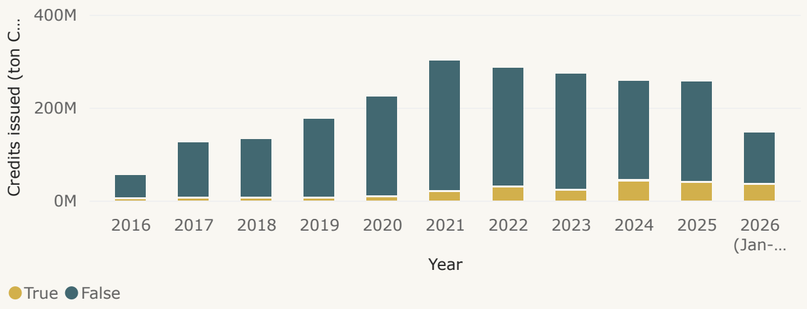
"We're trying to cut through all the low-quality credits that were issued in the past, which were meant mostly to catalyze the market. And now it's more about quality."
(NGO Certifier 1)

2.https://www.theguardian.com/environment/2023/jan/18/revealed-forest-carbon-offsets-biggest-provider-worthless-verra-aoe.

How are certification labels used?

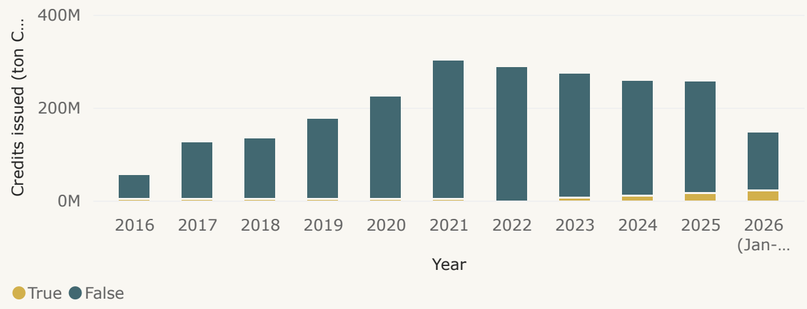
ICVCM CCP

ICVCM-CCP-Approved



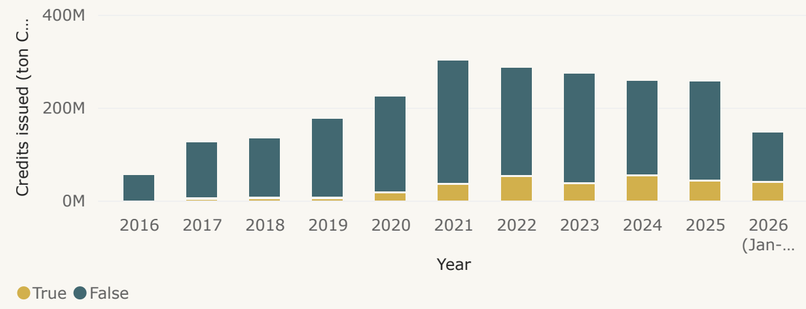
Article 6

Article-6-Authorized



CORSIA

CORSIA-Eligible



Year

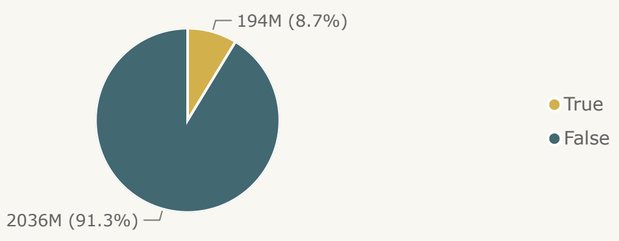
2016 — 2026

Carbon Standard

ACR | CAR | Gold Standard | Verra

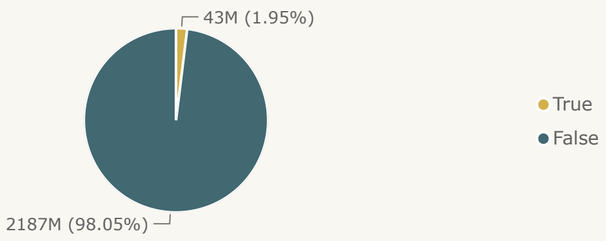
ICVCM-CCP-Approved

Credits issued (ton CO2e)



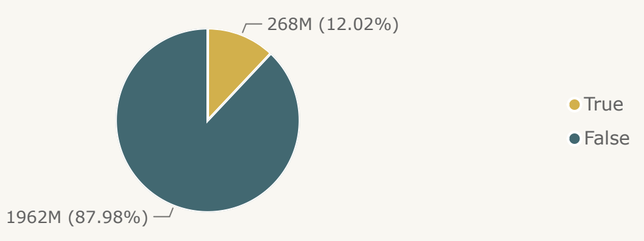
Article-6-Authorized

Credits issued (ton CO2e)



CORSIA-Eligible

Credits issued (ton CO2e)



ICVCM and CORSIA are more influential than Article 6
(NGO Watchdog 2, NGO Supplier 1, Business Supplier & Intermediary 1)

Key insights:

- **An increasing number of credits have been labelled** as CCP-approved, Article-6-authorized, and/or CORSIA-eligible, reflecting the transition **from credit quantity to quality**.
- **Article 6 lags behind ICVCM CCP and CORSIA**. Gold Standard, who has commited to align with the Paris Agreement[2], has fewer than 1% of its issued credits authorized by Article 6. Meanwhile, ACR and CAR have no Article 6-authorized credits. They, however, issued more CORSIA-eligible and CCP-approved credits than Gold Standard and Verra.

2.NGO Certifier 1; https://www.goldstandard.org/publications/paris-agreement-alignment-information.

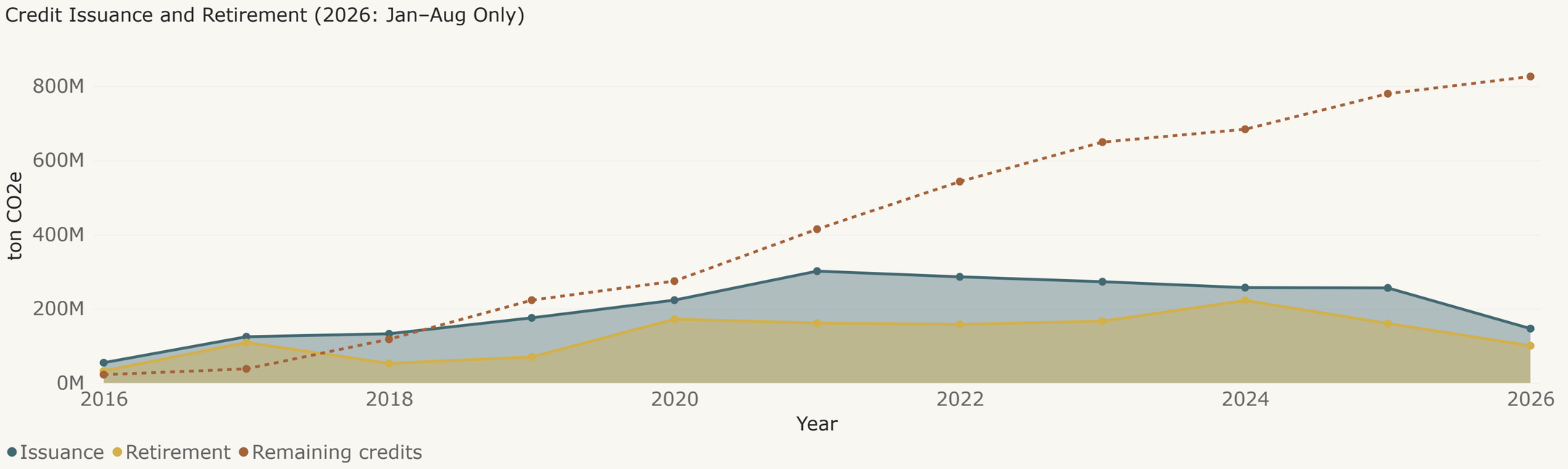
What's the overall trend over the last 10 years?

After issuance, carbon credits can be purchased by companies and eventually retired, typically to inset or offset their carbon emissions. In other words, **issuance** marks credits **entering the carbon market**, while **retirement** marks their **removal from circulation**.

Total issuance (ton CO2e)
2230M

Total retirement (ton CO2e)
1404M

Remaining credits (ton CO2e)
826M



Region: All | Reduction / Removal: All | Project Category: All | Carbon Standard: All | ICVCM CCP: All | Article 6: All | CORSIA: All

Discussion on the supply side shifted from urging buyers to pay more to cost-down because
"*people [suppliers] are just tired [of] requesting buyers to react*"
(NGO Certifier 4)

"*It's really a limited budget [for carbon credits] for now. That's because there are no mandatory requirements, not written by SBTi [Science-Based Targets initiative] or anything*"
(Business Buyer 1)

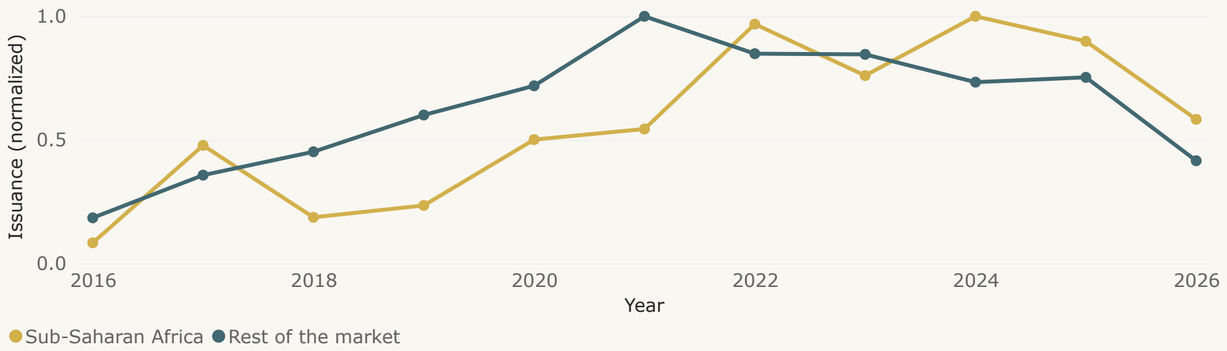
Key insights:

- Credit issuance consistently exceeds retirement, resulting in **a growing inventory of unretired credits**. This pattern is observed across multiple regions, project categories, carbon standards, and certification labels.
- Issuance and retirement generally increased in the early 2020s, followed by a gradual decline. Similiar patterns are observed across various market segments, suggesting the influence of broader market dynamics like the previously mentioned project quality challenge. However, issuance exhibits more pronounced changes, while **retirement remains relatively stable**, reflecting what expert interviews reveal: **There are currently no strong signals of growing demand**.

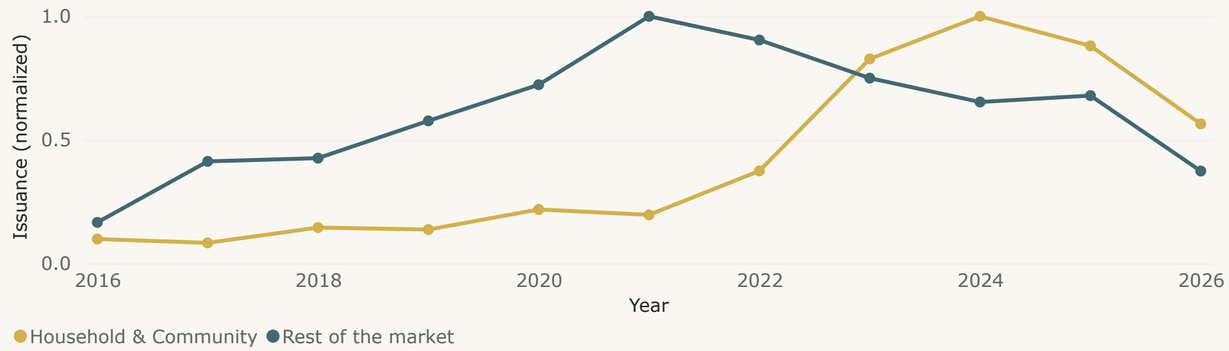
Which parts of the market are growing more compared to the overall market?

Various market segments are compared with the overall market on a normalized scale to identify which segments are growing.
2026 data only covers Jan-Aug.

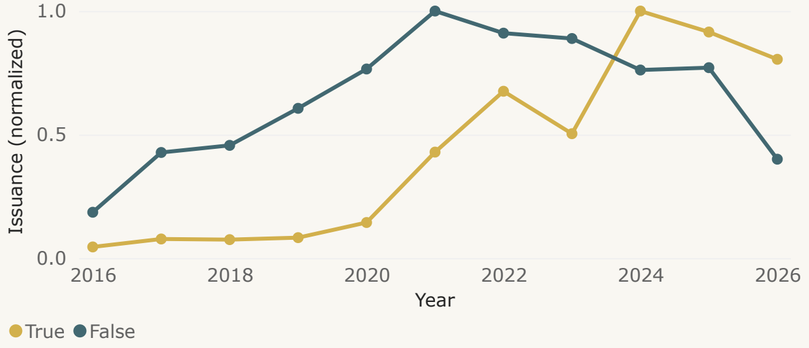
Region



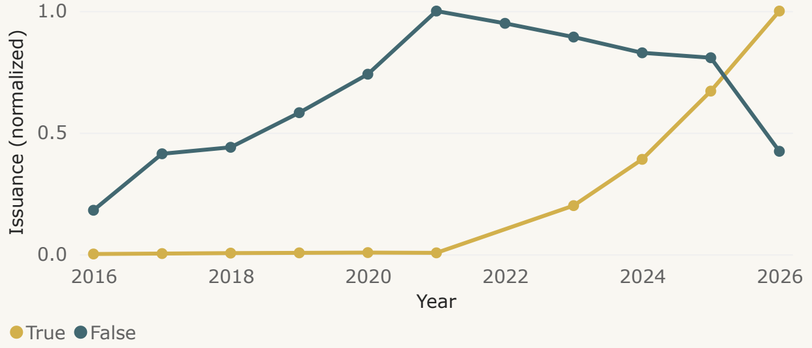
Project Category



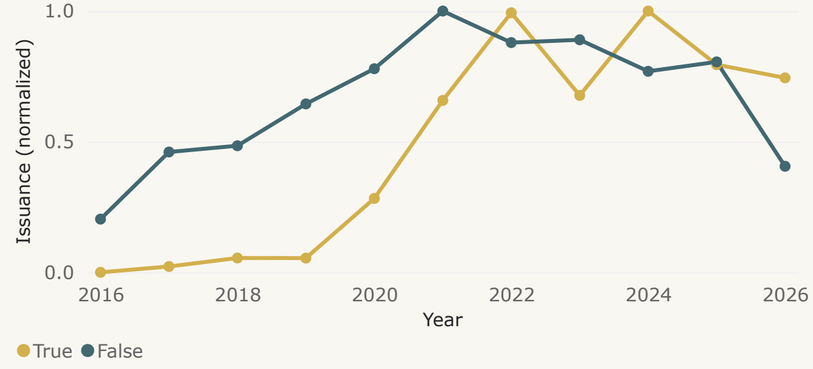
ICVCM-CCP-Approved



Article-6-Authorized



CORSIA-Eligible



There is a growing demand and price premium for credits generated from CCP-approved or CORSIA-eligile methodologies.
(NGO Certifier 4, NGO Certifier 4, NGO Watchdog 3, Policy Expert 1)

Business Supplier 3 is transitioning from developing own methodologies to using existing, CCP-approved methodologies.

Key insights:

Although there are currently no strong signals of growing demand, **some market segments are growing:**

- **Sub-Saharan African** projects
- **Household & community** (mostly cookstove) projects
- Projects with **CPP-approved**, **Article-6-authorized**, or **CORSIA-eligible** labels.

What are the future developments?

In April 2026, reports emerged that Microsoft would pause its purchases of carbon removal credits.

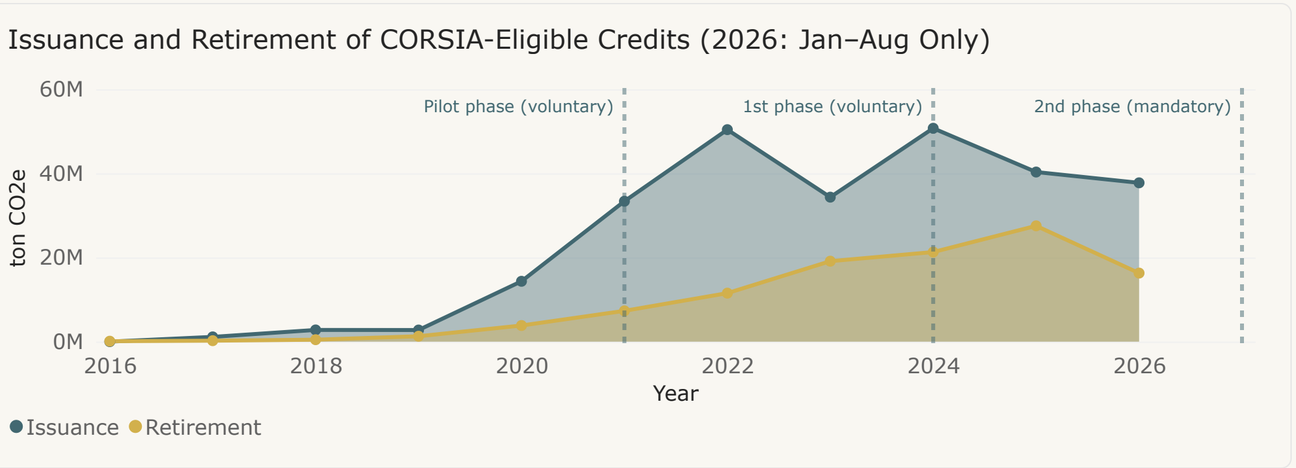
Why is Microsoft important?
Microsoft has been one of the biggest buyers who led the market.

> Voluntary carbon market heavily rely on a few buyers, including Microsoft,
> who act as first movers and build infrastructure.
> (NGO Certifier 4, Business Supplier 1, Business Supplier 3)

> *"If it hadn't been for Stripe, Frontier, Shopify, Google, and Microsoft,*
> *the CDR [carbon removal] ecosystem wouldn't exist."*
> (Business Intermediary 1)

> *"It is absolutely needed to bring more buyers in this market...if Microsoft is responsible for 90% of removal credits, that is not a market; That is a contract."*
> (Business Intermediary 1)

Where can the market find more buyers?
CORSIA might demonstrate how **compliance regulations can drive demand for carbon credits**, as the issuance and retirement of CORSIA-eligible credits have been growing.
In 2027, CORSIA will transition from the 1st phase to the 2nd phase. International airlines will be required to offset their emissions above the 2019 baselines on routes between participating countries. The key difference is that participation will become mandatory for most large economies, including Brazil, China, and India, which will join CORSIA in 2027[3].



> *"CORSIA credits are the ones that are most sought after...*
> *because they are going to give project developers highest premium,*
> *so a lot of market players are involved in them as investors, traders, project developers, and countries."*
> (NGO Certifier 2)

> *"We projected that CORSIA will require about 180 million credits in the first phase...the deadline for that is obviously the end of next year...*
> *you're gonna see a flurry of transactions happening for CORSIA between now [Apr 2026] and next year"*
> (Policy Expert 1)

Other regulatory developments that may drive demand include:
the proposal to integrate EU CRCF into EU ETS, the potential link between CBAM and Article 6 credits, and the develoment of national carbon markets

3.https://www.icao.int/sites/default/files/environmental-protection/Assembly42/Resolution-A42-22_CORSIA.pdf; https://aviationbenefits.org/other-environmental-challenges/climate-action/market-based-measures/corsia/corsia-explained/.

Takeaway

- **Forestry projects, including REDD+, in Latin America have lost momentum.** Once a major source of credits, the segment has declined noticeably since 2023.
- **Cookstove projects in Sub-Saharan Africa stand out as a potential growth segment.** Gold Standard has established an early lead over other major standards in this area
- **The project quality controversies have driven a transition from quantity to quality**.
- **Additional certification labels are supporting this transition.** ICVCM-CCP-approved, Article-6-authorized, and CORSIA-eligible labels still represent a relatively small share of the market (particularly Article 6), but their presence is steadily growing.
- **The transition from quantity to quality also coincides with a shift from voluntary to compliance-driven demand.** While some major voluntary buyers are slowing their purchases, CORSIA is expected to bring more international airlines into the market as its coverage expands.

Got questions or feedback?
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